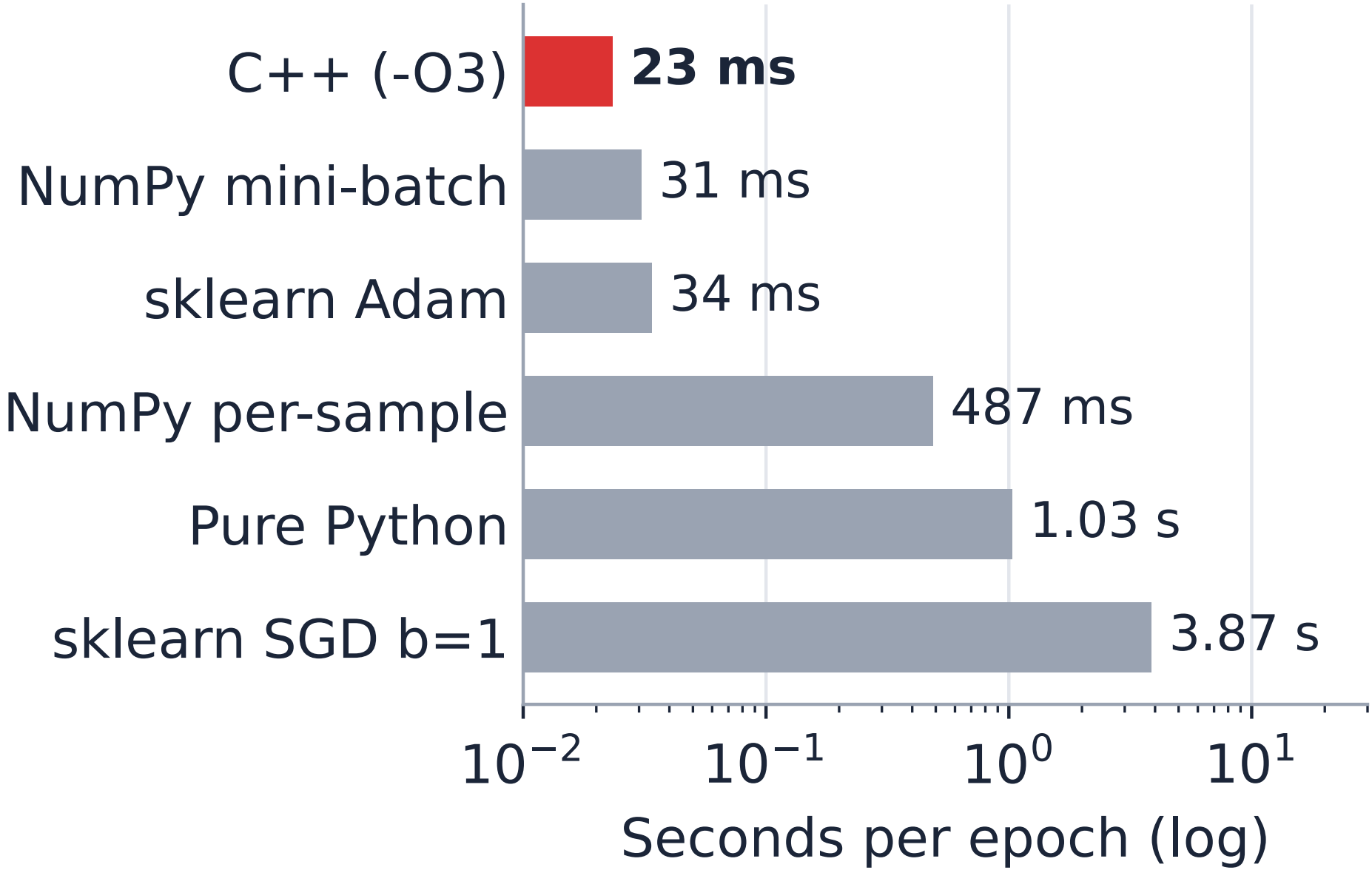
C++ (-O3)
23 ms
NumPy mini-batch
31 ms
sklearn Adam
34 ms
NumPy per-sample
487 ms
Pure Python
1.03 s
sklearn SGD b=1
3.87 s
10^{-2}
10^{-1}
10^{0}
10^{1}
Seconds per epoch (log)